Navigating the evolving landscape of **wildfire** management: A systematic **review** of decision support tools

T O'Mara, AS Meador, M Colavito, A Waltz… - Trees, Forests and …, 2024 - Elsevier

… to explore the landscape of decision support tools (DSTs) for **wildfire** management, … **review** led us to group decision support tools into categories for all DSTs, such as, Fire **Behavior** …

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Analysis of **wildfire** visualization systems for research and training: are they up for the challenge of the current state of wildfires?

CAT Cortes, S Thurow, A Ong… - IEEE transactions on …, 2023 - ieeexplore.ieee.org

… , this **survey** reviews various studies that use immersive and non-immersive data visualization techniques to depict **wildfire behavior** and … conditions on **simulation** of **wildfire spread**,” …

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Studying **wildfire behavior** using FIRETEC

R Linn, J Reisner, JJ Colman… - … journal of **wildland fire**, 2002 - CSIRO Publishing

… capacity for a physics-based **model** capable of describing **wildfire behavior** in a wide variety of circumstances. We believe that a three-dimensional **wildfire model** of this type could help …

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Toward a comprehensive **wildfire spread model**

RO Weber - International Journal of **Wildland Fire**, 1991 - CSIRO Publishing

… It is hoped that the presented ideas will stimulate discussion on the likely viability of a future comprehensive physical **model** of **wildfire spread**. This author believes the simple models …

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Role of flow topology in wind-driven **wildfire propagation**

S Viknesh, A Tohidi, F Afghah, R Stoll, A Arzani - Physics of Fluids, 2025 - pubs.aip.org

… In a comprehensive **review**, Sullivan (2009a) detailed the development of physics-based … Gaining such fundamental insights is crucial, as it enables the **prediction** of **wildfire spread** …

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Modeling **wildfire spread** with an irregular graph network

W Jiang, F Wang, G Su, X Li, G Wang, X Zheng… - Fire, 2022 - mdpi.com

… In this paper, we propose a **wildfire spread model** with an irregular graph … **wildfire spread model** with an irregular graph network (IGN) to simulate and analyze the spatiotemporal **spread** …

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Forest fire management and fire suppression strategies: a systematic literature **review**

B Tezcan, T Eren - Natural Hazards, 2025 - Springer

… and preventing the **spread** of fires. The combination of **simulation** and optimization techniques … This study provides a comprehensive **review** of mathematical models for **forest fire** …

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Forecasting daily **wildfire** activity using poisson regression

CA Graff, SR Coffield, Y Chen… - … on Geoscience and …, 2020 - ieeexplore.ieee.org

… We find that an intercept-only **prediction model** can capture this expected **behavior** and that predicting this overall mean trend alone is sufficient to improve on the predictive accuracy of …

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Data-driven **wildfire spread** modeling of european wildfires using a spatiotemporal graph neural network

M Rösch, M Nolde, T Ullmann, T Riedlinger - Fire, 2024 - mdpi.com

… yet applied STGNNs to **model** the daily **wildfire spread** on a … DL-based **wildfire spread model** to predict the daily **spread** of … uses a STGNN to **model** the **wildfire spread** for the first time. …

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Climate change impact on future **wildfire** danger and activity in southern Europe: a **review**

J Dupuy, H Fargeon, N Martin-StPaul, F Pimont… - Annals of Forest …, 2020 - Springer

… moisture, which in turn conditions fire **spread** in the fire module. Both LPJ-… **spread** in SPITFIRE, while the maximum daily Nesterov index is used to calibrate the annual burnt area **model** …

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